Dokumen Scope, Pembagian Peran Tim

Kelompok 3

Pertemuan 1 — Kickoff Case 1: Vulnerability OJS

Tujuan Pembelajaran

Setelah pertemuan ini, mahasiswa mampu:

1. Mendefinisikan ruang lingkup (scope) vulnerability assessment secara profesional
2. Membentuk tim penguji dengan peran yang jelas
3. Menyiapkan lingkungan VPS dan menginstal OJS versi vulnerable
4. Memahami Rules of Engagement (RoE) dalam pengujian keamanan

---

1. Latar Belakang

Open Journal Systems (OJS) adalah platform manajemen dan penerbitan jurnal ilmiah open-source yang dikembangkan oleh Public Knowledge Project (PKP). OJS banyak digunakan oleh institusi akademik di seluruh dunia, termasuk perguruan tinggi di Indonesia.

Karena popularitasnya, OJS menjadi target menarik bagi penyerang. Beberapa versi OJS lama memiliki kerentanan kritis yang terdokumentasi di database CVE, antara lain:

CVE	Versi Terdampak	Jenis Kerentanan	Severity
CVE-2020-28112	< 3.2.1-2	Stored XSS	High

CVE-2021-27188	≤ 3.3.0-6	Server-Side Request Forgery (SSRF)	High
CVE-2022-24822	≤ 3.3.0-8	Open Redirect + XSS	Medium
CVE-2023-28131	< 3.3.0-12	XSS via Editor	High
CVE-2020-12432	2.x	SQL Injection	Critical

Catatan Etika: Seluruh pengujian dilakukan di lingkungan yang telah disetujui (VPS milik kelas). Dilarang keras menguji sistem di luar scope yang ditetapkan.

---

2. Pembentukan Tim

kelompok terdiri dari 8 mahasiswa dengan pembagian peran sebagai berikut:

Peran	Jumlah	Tanggung Jawab
Project Lead / Scrum Master	1	Koordinasi jadwal, komunikasi, final review laporan
Security Engineer (DAST)	3	Menjalankan scanning aktif (ZAP, Nikto, SQLMap)
Security Engineer (SAST)	2	Analisis source code statis (Semgrep, phpcs)
Documentation & Reporting	2	Menulis laporan, risk register, rekomendasi mitigasi

Daftar Anggota Tim

Nama Tim : Kelompok 3
Kelas : DevSecOps-A

No	Nama	NIM	Peran	TTD
1	Joshua Washington Hutasoit	235150207111037	Security Engineer (DAST)	
2	Brahmantio Jati Pambudi	235150201111028	Documentation & Reporting	
3	Malvinshah Haris Athala	235150207111042	Security Engineer (DAST)	
4	Steven Anthony	235150201111035	Security Engineer (DAST)	
5	Darvesh Aziz Mawla	235150201111036	Security Engineer (SAST)	
6	Sandhika Rizqi Ramadhan	235150200111019	Project Lead	
7	Rhesa Tsaqif Adyatma	235150200111020	Documentation & Reporting	
8	Mohamad Robi Alwan	235150200111025	Security Engineer (SAST)	

Repositori Tim (GitHub) : https://github.com/DarveshAziz/devsecops-caseOJS-kelompok3

---

3. Scope & Rules of Engagement (RoE)

3.1 In-Scope

- Aplikasi OJS yang berjalan di VPS yang diberikan dosen
- URL target: `http://10.34.100.180/`
- Port yang diizinkan untuk diuji: 80, 443, 8080
- Semua fitur OJS yang dapat diakses via browser (sebagai guest, author, reviewer, editor)

3.2 Out-of-Scope

- Infrastruktur jaringan kampus
- VPS tim lain
- Server PKP (upstream OJS)
- Pengujian DoS / DDoS (dilarang keras)
- Eksploitasi di luar kerentanan yang ditemukan (pivoting ke server lain)

3.3 Rules of Engagement

Aturan	Keterangan
Waktu pengujian	selama waktu penugasan
Metode	Non-destructive (jangan hapus/ubah data produksi)
Dokumentasi	Semua aktivitas harus dicatat (log, screenshot)
Pelaporan	Temuan kritis wajib dilaporkan ke dosen dalam 24 jam
Etika	Tidak membagikan akses VPS ke pihak luar

---

4. Persiapan Lingkungan VPS

4.1 Spesifikasi VPS yang Diberikan

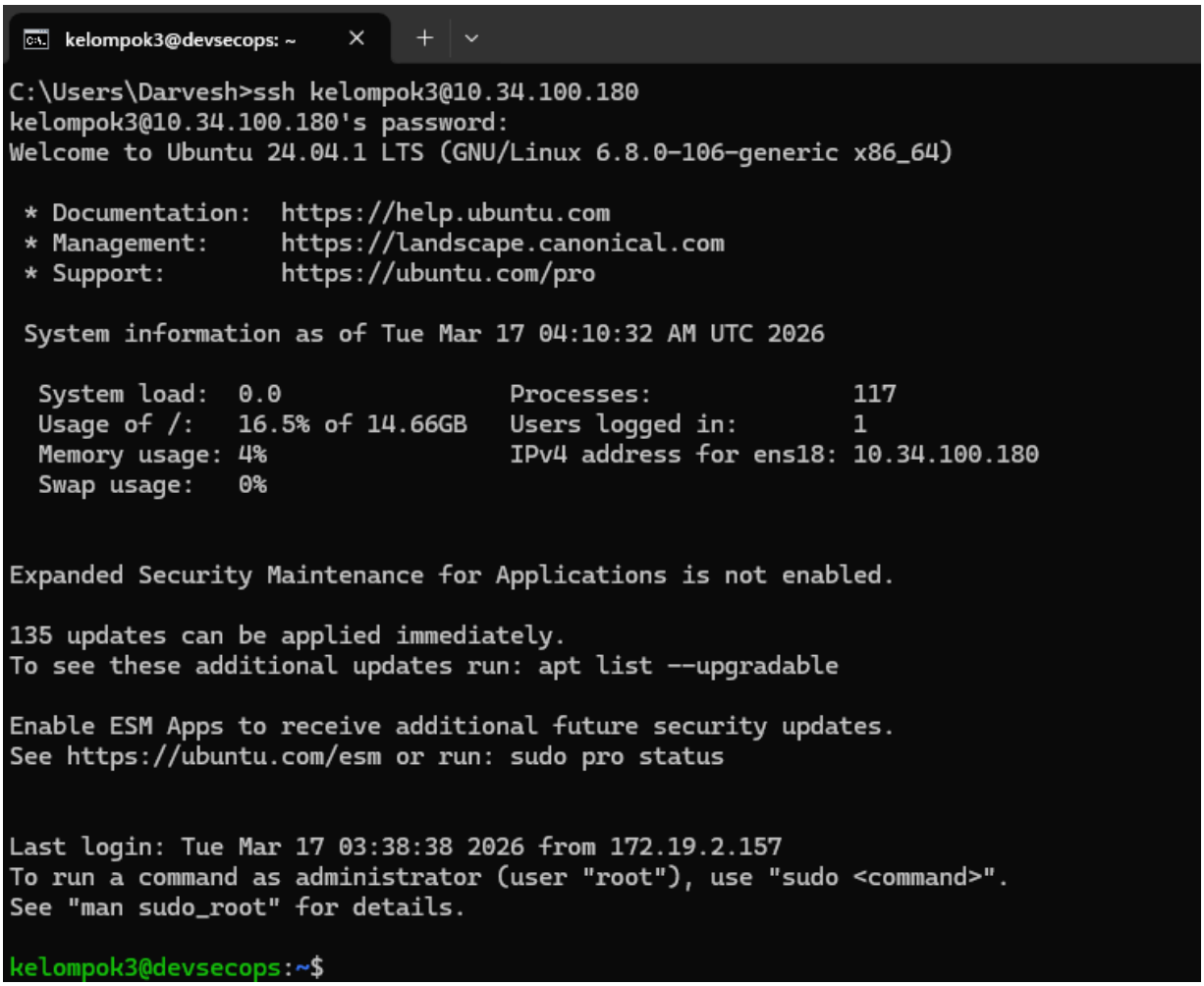
```
OS       : Ubuntu 22.04 LTS
CPU      : 2 vCPU
RAM      : 8 GB
Storage  : 32 GB SSD
```

4.2 Instalasi OJS Versi Vulnerable

Target versi: OJS 3.3.0-8 (mengandung CVE-2021-27188 dan CVE-2022-24822) atau bisa versi yang lain

Langkah 1 — Akses VPS via SSH

```
ssh ubuntu@10.34.100.180
```



Langkah 2 — Instalasi dependensi

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y apache2 mysql-server php7.4 php7.4-cli \
  php7.4-common php7.4-mysql php7.4-xml php7.4-mbstring \
  php7.4-gd php7.4-curl php7.4-zip php7.4-intl unzip wget
```

```
sudo apt update && sudo apt upgrade -y
```

```
kelompok3@devsecops: ~
kelompok3@devsecops:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for kelompok3:
Hit:1 http://id.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://id.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://id.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,540 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [976 kB]
Fetched 2,643 kB in 4s (714 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
134 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
  python3-boto3 python3-botocore python3-dateutil python3-jmespath python3-packaging python3-s3transfer
The following packages will be upgraded:
  apparmor apt apt-utils base-files cloud-init cloud-initramfs-copymods cloud-initramfs-dyn-netconf coreutils
  cryptsetup cryptsetup-bin cryptsetup-initramfs dhcpcd-base distro-info-data dmeventd dmidecode dmsetup
  dracut-install fwupd gir1.2-packagekitglib-1.0 gzip ibverbs-providers initramfs-tools initramfs-tools-bin
  initramfs-tools-core iproute2 kmod kpartx landscape-common libacl1 libaio1t64 libapparmor1 libapt-pkg6.0t64 libattr1
  libaudit-common libaudit1 libbsd0 libcryptsetup12 libdevmapper-event1.02.1 libdevmapper1.02.1 libdrm-common libdrm2
  libfwupd2 libgmp10 libgpg-error-l10n libgpg-error0 libibverbs1 libidn2-0 libkmod2 libldap-common libldap2
  liblvm2cmd2.03 libmbim-glib4 libmbim-proxy libmbim-utils libmd0 libmpfr6 libnetplan1 libnghttp2-14 libnl-3-200
  libnl-genl-3-200 libnl-route-3-200 libnss-systemd libnuma1 libnvme1t64 libopeniscsiusr libpackagekit-glib2-18
  libpam-systemd libpcre2-8-0 libplymouth5 libpolkit-agent-1-0 libpolkit-gobject-1-0 libproc2-0 libselinux1
  libsystemd-shared libsystemd0 libtraceevent1 libtraceevent1-plugin libudev1 libunistring5 libunwind8 linux-base
  login lvm2 lxd-agent-loader mdadm motd-news-config mtr-tiny multipath-tools netplan-generator netplan.io numactl
  open-iscsi overlayroot packagekit packagekit-tools passwd pci.ids plymouth plymouth-theme-ubuntu-text polkitd
  pollinate powermgmt-base procps python-apt-common python3-apt python3-distupgrade python3-netplan
  python3-software-properties python3-update-manager rsyslog snapd software-properties-common sosreport systemd
  systemd-dev systemd-hwe-hwdb systemd-resolved systemd-sysv systemd-timesyncd tcpdump tmux ubuntu-drivers-common
  ubuntu-pro-client ubuntu-pro-client-l10n ubuntu-release-upgrader-core udev update-manager-core
  update-notifier-common vim vim-common vim-runtime vim-tiny xfsprogs xxd
134 upgraded, 6 newly installed, 0 to remove and 0 not upgraded.
5 standard LTS security updates
Need to get 87.2 MB of archives.
After this operation, 115 MB of additional disk space will be used.
Get:1 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 motd-news-config all 13ubuntu10.4 [4,012 B]
Get:2 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 base-files amd64 13ubuntu10.4 [73.3 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/main amd64 vim amd64 2:9.1.0016-1ubuntu7.10 [1,882 kB]
Get:4 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 coreutils amd64 9.4-3ubuntu6.1 [1,413 kB]
Get:5 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 gzip amd64 1.12-1ubuntu3.1 [99.0 kB]
Get:6 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 login amd64 1:4.13+dfsg1-4ubuntu3.2 [202 kB]
Get:7 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-timesyncd amd64 255.4-1ubuntu8.12 [35.3 kB]
Get:8 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-resolved amd64 255.4-1ubuntu8.12 [296 kB]
Get:9 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 libacl1 amd64 2.3.2-1build1.1 [16.8 kB]
Get:10 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapparmor1 amd64 4.0.1really4.0.1-0ubuntu0.24.04.5 [50.5 kB]
```

```
kelompok3@devsecops: ~
Setting up cloud-initramfs-dyn-netconf (0.49~24.04.1) ...
Setting up overlayroot (0.49~24.04.1) ...
Setting up cryptsetup-initramfs (2:2.7.0-1ubuntu4.2) ...
update-initramfs: deferring update (trigger activated)
Setting up liblvm2cmd2.03:amd64 (2.03.16-3ubuntu3.2) ...
Setting up dmeventd (2:1.02.185-3ubuntu3.2) ...
dm-event.service is a disabled or a static unit not running, not starting it.
Setting up python3-update-manager (1:24.04.12) ...
Setting up python3-distupgrade (1:24.04.28) ...
Setting up lvm2 (2.03.16-3ubuntu3.2) ...
Setting up ubuntu-release-upgrader-core (1:24.04.28) ...
Setting up update-manager-core (1:24.04.12) ...
Setting up update-notifier-common (3.192.68.2) ...
update-notifier-download.service is a disabled or a static unit not running, not starting it.
update-notifier-motd.service is a disabled or a static unit not running, not starting it.
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for sgml-base (1.31) ...
Processing triggers for debianutils (5.17build1) ...
Processing triggers for install-info (7.1-3build2) ...
Setting up packagekit (1.2.8-2ubuntu1.4) ...
Setting up packagekit-tools (1.2.8-2ubuntu1.4) ...
Setting up software-properties-common (0.99.49.3) ...
Processing triggers for initramfs-tools (0.142ubuntu25.8) ...
update-initramfs: Generating /boot/initrd.img-6.8.0-106-generic
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
 /etc/needrestart/restart.d/systemd-manager
 systemctl restart cron.service ssh.service systemd-journald.service systemd-networkd.service udisks2.service upower.service

Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 kelompok3 @ session #12: sshd[1430]
 kelompok3 @ session #18: apt[2044], sshd[1625]
 kelompok3 @ session #4: sshd[1115]
 kelompok3 @ user manager service: systemd[1120]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
kelompok3@devsecops:~$
```

```
sudo apt install -y apache2 mysql-server php7.4 php7.4-cli \
  php7.4-common php7.4-mysql php7.4-xml php7.4-mbstring \
  php7.4-gd php7.4-curl php7.4-zip php7.4-intl unzip wget
```

```
kelompok3@devsecops: ~
kelompok3@devsecops:~$ sudo apt install -y apache2 mysql-server php7.4 php7.4-cli \
  php7.4-common php7.4-mysql php7.4-xml php7.4-mbstring \
  php7.4-gd php7.4-curl php7.4-zip php7.4-intl unzip wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapache2-mod-php7.4 libaprlt64 libaprutil1-dbd-sqlite3 libaprutil1-ldap
  libaprutil1t64 libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64
  libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl
  libhttp-message-perl libio-html-perl liblua5.4-0 liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64
  libtimedate-perl liburi-perl libzip4t64 mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0
  mysql-client-core-8.0 mysql-common mysql-server-8.0 mysql-server-core-8.0 php-common php7.4-json php7.4-opcache
  php7.4-readline ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser php-pear libdata-dump-perl
  libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl libregexp-ipv6-perl libwww-perl mailx
  tinyca zip
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapache2-mod-php7.4 libaprlt64 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libaprutil1t64 libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl
  libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl
  libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblua5.4-0 liblwp-mediatypes-perl
  libmecab2 libprotobuf-lite32t64 libtimedate-perl liburi-perl libzip4t64 mecab-ipadic mecab-ipadic-utf8 mecab-utils
  mysql-client-8.0 mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0 mysql-server-core-8.0 php-common
  php7.4 php7.4-cli php7.4-common php7.4-curl php7.4-gd php7.4-intl php7.4-json php7.4-mbstring php7.4-mysql
  php7.4-opcache php7.4-readline php7.4-xml php7.4-zip ssl-cert unzip
0 upgraded, 55 newly installed, 0 to remove and 1 not upgraded.
Need to get 37.0 MB of archives.
After this operation, 272 MB of additional disk space will be used.
Get:1 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 libaprlt64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:4 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9,116 B]
Get:5 http://id.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://id.archive.ubuntu.com/ubuntu noble/main amd64 mysql-common all 5.8+1.1.0build1 [6,746 B]
Get:7 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-core-8.0 amd64 8.0.45-0ubuntu0.24.04.1 [2,740 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.11 [1,335 kB]
Get:9 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-8.0 amd64 8.0.45-0ubuntu0.24.04.1 [22.5 kB]
Get:10 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libevent-pthreads-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [7,982 B]
Get:11 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libmecab2 amd64 0.996-14ubuntu4 [201 kB]
Get:12 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-server-core-8.0 amd64 8.0.45-0ubuntu0.24.04.1 [17.6 MB]
Get:13 https://ppa.launchpadcontent.net/ondrej/php/ubuntu noble/main amd64 php-common all 2:101~+ubuntu24.04.1+deb.sury.org+1 [14.2 kB]
Get:14 https://ppa.launchpadcontent.net/ondrej/php/ubuntu noble/main amd64 php7.4-common amd64 1:7.4.33-26+ubuntu24.04.1+deb.sury.org+1 [688 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/main amd64 apache2-data all 2.4.58-1ubuntu8.11 [163 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.11 [98.2 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/main amd64 apache2 amd64 2.4.58-1ubuntu8.11 [90.2 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/main amd64 libprotobuf-lite32t64 amd64 3.21.12-8.2ubuntu0.3 [238 kB]
Get:19 https://ppa.launchpadcontent.net/ondrej/php/ubuntu noble/main amd64 php7.4-json amd64 1:7.4.33-26+ubuntu24.04.1+deb.sury.org+1 [20.6 kB]
```

```
kelompok3@devsecops: ~
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
info: Switch to mpm prefork for package libapache2-mod-php7.4
Module mpm_event disabled.
Enabling module mpm_prefork.
info: Executing deferred 'a2enmod php7.4' for package libapache2-mod-php7.4
Enabling module php7.4.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Setting up php7.4 (1:7.4.33-26+ubuntu24.04.1+deb.sury.org+1) ...
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for php7.4-cli (1:7.4.33-26+ubuntu24.04.1+deb.sury.org+1) ...
Processing triggers for libapache2-mod-php7.4 (1:7.4.33-26+ubuntu24.04.1+deb.sury.org+1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 kelompok3 @ session #12: sshd[1430]
 kelompok3 @ session #18: sshd[1625]
 kelompok3 @ session #4: sshd[1115]
 kelompok3 @ user manager service: systemd[1120]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
kelompok3@devsecops:~$
```

Langkah 3 — Unduh OJS versi vulnerable

```
cd /var/www/html
sudo wget https://pkp.sfu.ca/ojs/download/ojs-3.3.0-8.tar.gz
sudo tar -xzf ojs-3.3.0-8.tar.gz
sudo mv ojs-3.3.0-8 ojs
sudo chown -R www-data:www-data ojs
sudo chmod -R 755 ojs
```

```
kelompok3@devsecops:~$ cd /var/www/html
kelompok3@devsecops:/var/www/html$ sudo wget https://pkp.sfu.ca/ojs/download/ojs-3.3.0-8.tar.gz
[sudo] password for kelompok3:
--2026-03-17 04:36:29--  https://pkp.sfu.ca/ojs/download/ojs-3.3.0-8.tar.gz
Resolving pkp.sfu.ca (pkp.sfu.ca)... 172.255.54.247
Connecting to pkp.sfu.ca (pkp.sfu.ca)|172.255.54.247|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 51182913 (49M) [application/x-gzip]
Saving to: 'ojs-3.3.0-8.tar.gz'

ojs-3.3.0-8.tar.gz                                  100%[=================================================

2026-03-17 04:36:41 (4.74 MB/s) - 'ojs-3.3.0-8.tar.gz' saved [51182913/51182913]

kelompok3@devsecops:/var/www/html$ sudo tar -xzf ojs-3.3.0-8.tar.gz
kelompok3@devsecops:/var/www/html$ sudo mv ojs-3.3.0-8 ojs
kelompok3@devsecops:/var/www/html$ sudo chown -R www-data:www-data ojs
kelompok3@devsecops:/var/www/html$ sudo chmod -R 755 ojs
kelompok3@devsecops:/var/www/html$
```

Langkah 4 — Buat database MySQL

```
sudo mysql -u root -e "
  CREATE DATABASE ojs_db CHARACTER SET utf8mb3;
  CREATE USER 'ojs_user'@'localhost' IDENTIFIED BY 'P@ssw0rd_OJS!';
  GRANT ALL PRIVILEGES ON ojs_db.* TO 'ojs_user'@'localhost';
  FLUSH PRIVILEGES;
"
```

```
kelompok3@devsecops:/$ sudo mysql -u root -e "
  CREATE DATABASE ojs_db CHARACTER SET utf8mb3;
  CREATE USER 'ojs_user'@'localhost' IDENTIFIED BY 'P@ssw0rd_OJS!';
  GRANT ALL PRIVILEGES ON ojs_db.* TO 'ojs_user'@'localhost';
  FLUSH PRIVILEGES;
"
kelompok3@devsecops:/$
```

Langkah 5 — Konfigurasi Apache Virtual Host

```
sudo nano /etc/apache2/sites-available/ojs.conf
```

Isi file konfigurasi:

```
<VirtualHost *:80>
    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html/ojs
```

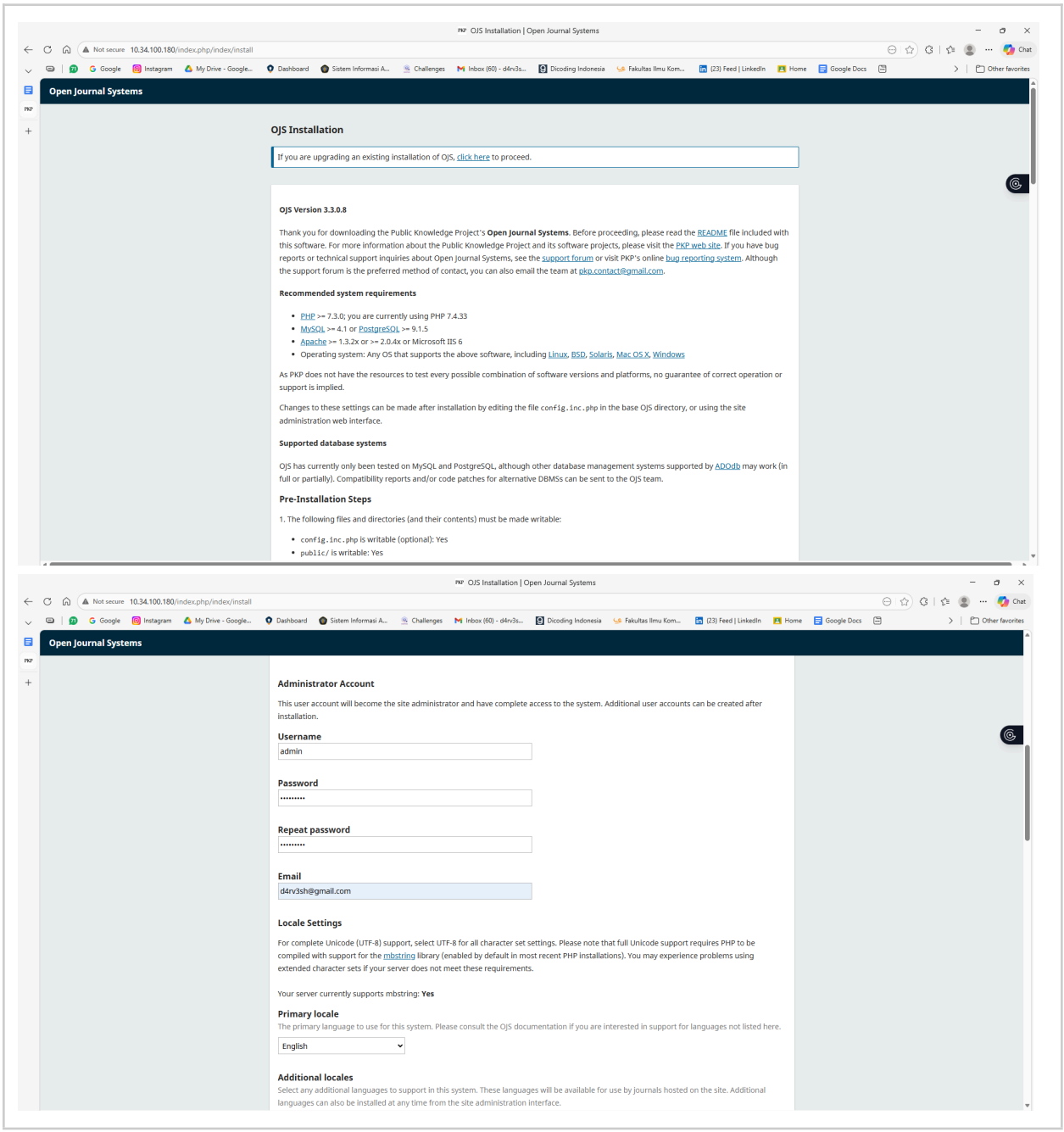
```
    ServerName 10.34.100.180

    <Directory /var/www/html/ojs>
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/ojs_error.log
    CustomLog ${APACHE_LOG_DIR}/ojs_access.log combined
</VirtualHost>

sudo a2ensite ojs.conf
sudo a2enmod rewrite
sudo systemctl restart apache2
```

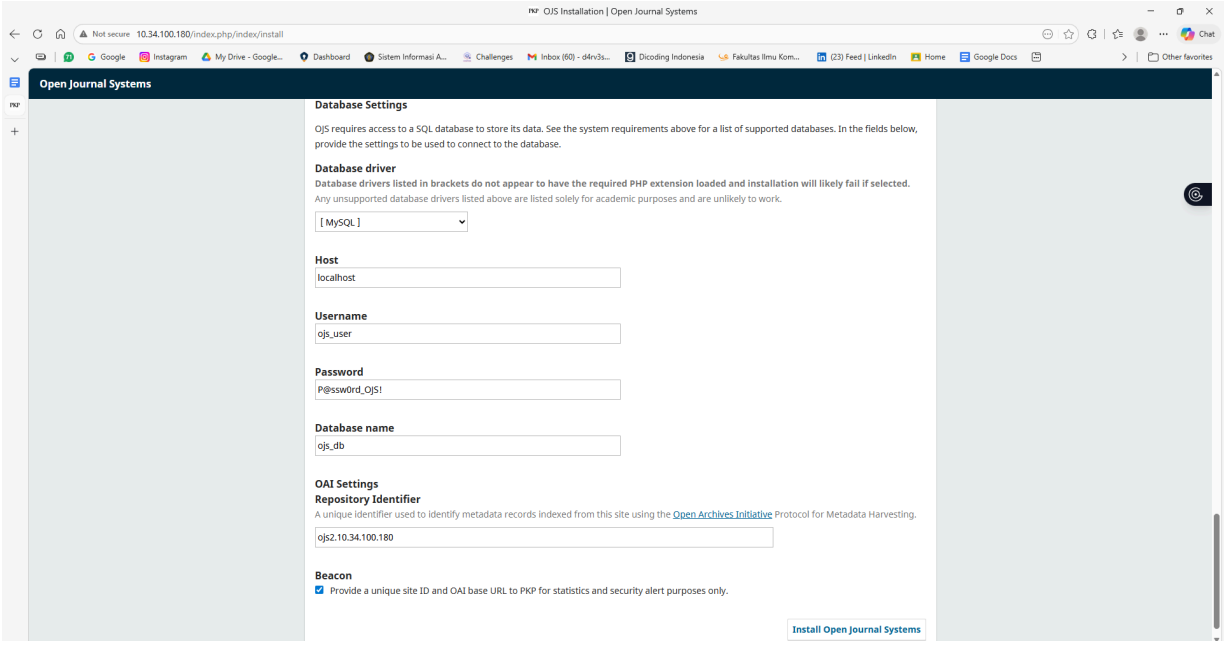
```
kelompok3@devsecops:/$ sudo nano /etc/apache2/sites-available/ojs.conf
kelompok3@devsecops:/$ sudo cat /etc/apache2/sites-available/ojs.conf
<VirtualHost *:80>
    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html/ojs
    ServerName 10.34.100.180

    <Directory /var/www/html/ojs>
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/ojs_error.log
    CustomLog ${APACHE_LOG_DIR}/ojs_access.log combined
</VirtualHost>
kelompok3@devsecops:/$ sudo a2ensite ojs.conf
Enabling site ojs.
To activate the new configuration, you need to run:
  systemctl reload apache2
kelompok3@devsecops:/$ sudo a2enmod rewrite
Enabling module rewrite.
To activate the new configuration, you need to run:
  systemctl restart apache2
kelompok3@devsecops:/$ sudo systemctl restart apache2
kelompok3@devsecops:/$
```

Langkah 6 — Selesaikan instalasi via browser

```
Buka browser → http://10.34.100.180/index.php/index/install
```

```
Ikuti wizard instalasi dengan data berikut:
  - Database Host     : localhost
  - Database Name     : ojs_db
  - Database User     : ojs_user
  - Database Password: P@ssw0rd_OJS!
  - Admin Username    : admin
  - Admin Password    : Admin@123 (catat dengan baik!)
```





Client character set

The encoding to use for data sent to and received from browsers.

Unicode (UTF-8)

Connection character set

Unicode (UTF-8)

File Settings

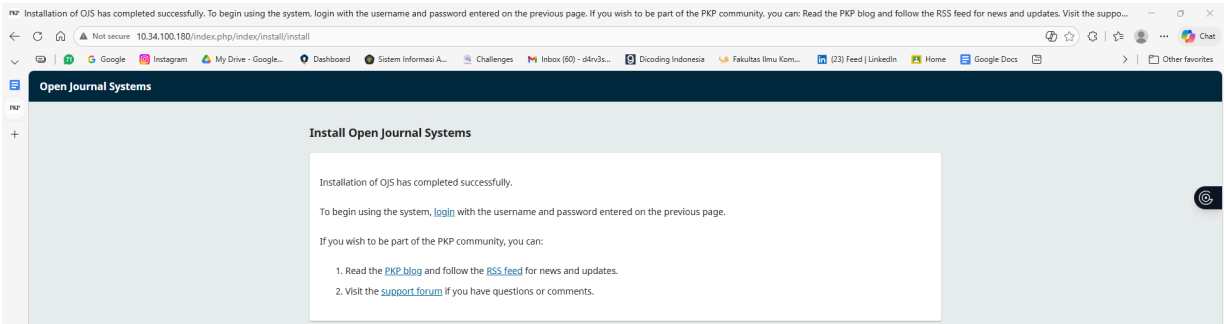
Directory for uploads

Enter full pathname to an existing directory where uploaded files are to be kept. This directory should not be directly web-accessible. **Please ensure that this directory exists and is writable prior to installation.** Windows path names should use forward slashes, e.g. "C:/myjournal/files".

/var/www/html/files

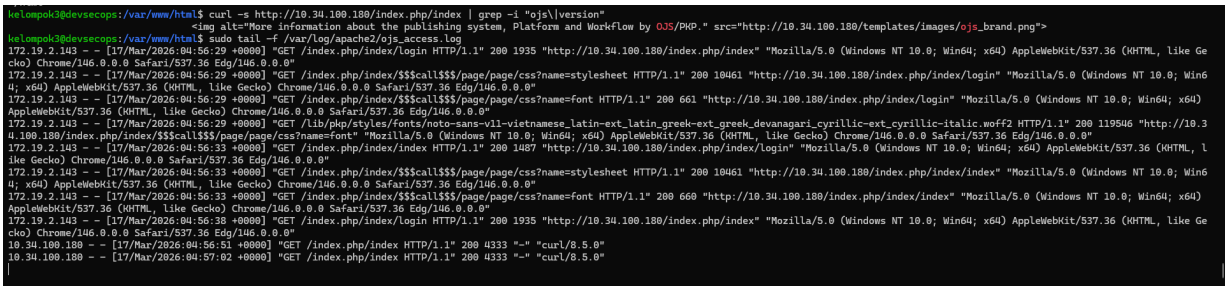
Your server currently allows file uploads: **Yes**

Your server currently allows a maximum file upload size of: **2M**



Langkah 7 — Verifikasi instalasi

```
# Cek versi OJS lewat terminal
curl -s http://10.34.100.180/index.php/index | grep -i "ojs\|version"

# Cek log Apache
sudo tail -f /var/log/apache2/ojs_access.log
```



---

5. Deliverable Pertemuan 1

No	Deliverable	Format	Dikumpulkan Via
1	Dokumen Scope & RoE (ditandatangani tim)	PDF	LMS / GitHub
2	Daftar anggota tim + peran	MD	GitHub repo tim
3	Screenshot OJS berhasil diinstal (menampilkan versi)	PNG	GitHub repo tim
4	URL VPS aktif	Plain text	LMS

---

6. Pertanyaan Diskusi

1. Mengapa penting mendefinisikan scope sebelum melakukan penetration testing?

   Pendefinisian scope sangat krusial karena berfungsi sebagai batasan legal dan teknis yang menjaga pengujian tetap pada jalurnya. Secara profesional, scope menentukan aset digital mana yang boleh disentuh, metode serangan yang diizinkan (misalnya: apakah boleh melakukan brute force atau tidak), serta durasi pengujian agar tidak berbenturan dengan waktu operasional puncak sistem.

   Tanpa scope yang jelas, tim penguji berisiko melakukan tindakan yang tidak terkontrol yang bisa menyentuh infrastruktur pihak ketiga (seperti cloud provider atau API luar) yang tidak memberikan izin, sehingga berpotensi menimbulkan jeratan hukum atau pelanggaran etika profesional. Selain itu, scope membantu memastikan bahwa sumber daya tim yang terbatas dialokasikan secara efektif untuk memprioritaskan aset yang paling kritis bagi organisasi.

2. Apa risiko yang mungkin terjadi jika pengujian dilakukan tanpa Rules of Engagement?

   Melakukan pengujian tanpa Rules of Engagement (RoE) membawa risiko tinggi secara teknis, organisasional, dan legal. Secara teknis, pengujian bisa menjadi tidak terkendali dan bersifat destruktif, seperti menyebabkan korupsi database, menghapus data secara permanen, atau memicu downtime yang melumpuhkan layanan.

   Dari sisi organisasional, ketiadaan RoE sering kali memicu konflik internal karena tidak adanya protokol komunikasi yang disepakati, seperti siapa yang harus dihubungi jika terjadi insiden darurat atau bagaimana prosedur eskalasi temuan kritis.

   Secara legal, tanpa RoE yang tertandatangan, seluruh aktivitas pengujian dapat dikategorikan sebagai akses ilegal atau serangan siber oleh pihak berwenang, karena tidak ada bukti tertulis mengenai persetujuan pemilik sistem terhadap metode pengujian yang dilakukan.

3. Jelaskan perbedaan antara vulnerability assessment dan penetration testing!

Vulnerability assessment berfokus pada identifikasi dan pemetaan kelemahan sebanyak mungkin (breadth), lalu memberi prioritas risiko untuk mitigasi. Proses ini biasanya menggunakan alat pemindai otomatis untuk menemukan celah keamanan yang ada di seluruh permukaan sistem. Output utamanya adalah daftar inventaris kerentanan lengkap dengan tingkat keparahan yang didasarkan pada skor risiko.

Penetration testing berfokus pada pembuktian eksploitasi kelemahan tertentu untuk melihat dampak nyata terhadap bisnis atau sistem (depth). Penguji akan mensimulasikan perilaku penyerang sungguhan untuk menembus pertahanan sistem dan melihat seberapa jauh akses yang bisa didapatkan. Output utamanya adalah bukti eksploitasi (proof of concept), jalur serangan yang berhasil dilewati, dampak nyata, dan rekomendasi perbaikan berbasis skenario serangan.

Singkatnya: assessment menjawab apa saja yang lemah, pentest menjawab seberapa jauh kelemahan itu bisa disalahgunakan.

4. Mengapa OJS versi lama masih banyak dipakai di institusi akademik? Apa implikasinya terhadap keamanan?

Alasan umum: keterbatasan anggaran atau SDM, ketergantungan pada plugin atau tema lama yang belum tentu kompatibel dengan versi terbaru, kekhawatiran akan kegagalan migrasi data yang dapat mengganggu operasional jurnal saat upgrade, dan proses birokrasi kampus yang lambat dalam menyetujui pemeliharaan infrastruktur IT.

Implikasi keamanannya: sistem menjadi jauh lebih rentan terhadap daftar CVE yang sudah diketahui oleh publik, sehingga lebih mudah dipindai dan dieksploitasi oleh penyerang menggunakan skrip otomatis. Hal ini berisiko tinggi pada kerahasiaan data pribadi pengguna, integritas naskah ilmiah atau proses review, serta ketersediaan layanan jurnal yang bisa lumpuh sewaktu-waktu akibat serangan siber.

Referensi

- OWASP Testing Guide v4.2 — Chapter: Test Management
- PTES (Penetration Testing Execution Standard) — Pre-engagement Interactions
- PKP OJS Security Advisories: https://pkp.sfu.ca/ojs/